The Augmented Synthetic Historical Control Method

1 Introduction

Causal inference in economic and policy research often compares a treated set of units to a control or donor pool. The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics, second only to difference-in-differences. The idea of SC is straightforward: the untreated potential outcomes of a treated unit are approximated by a weighted average of donor units. Difference-in-differences relies on the assumption of parallel pre-intervention trends [Baker et al., 2025].

In many settings, however, no suitable donor units are available—for example, during global events such as pandemics or financial crises, or when no valid donor units exist. When no valid untreated donor units are available, researchers often turn to the interrupted time series (ITS) design. In its canonical form, ITS models the pre-treatment outcome path as a parametric linear trend and tests for changes in level or slope following an intervention. In fact, even comparative interrupted time series designs have been used by policy analysts [Coopersmith et al., 2022]. Even so, ITS can be fragile in many realistic applications. Its validity depends critically on the assumption that an OLS line fitted to the pre-treatment period provides a reasonable counterfactual projection. In practice, many outcome series feature nonlinear dynamics, seasonality, or structural breaks that violate this assumption. Moreover, ITS typically imposes a homogeneous effect structure, assuming a single jump in level and/or slope that applies equally across all post-treatment periods. These restrictions can lead to biased inference when the true effects are time-varying or when pre-treatment is poor.

To address this shortcoming, Chen et al. [2024] proposed the synthetic historical control method (SHC). Rather than relying on external donors, SHC constructs a donor pool from the treated unit's own pre-treatment history, splitting the pre-treatment period into overlapping segments. The SHC also, unlike ITS, uses kernel smoothing to account for non-linearities in the pre-treatment time series data which ITS does not naturally account for. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, allowing for effect size heterogeneity found in SCM and increasingly DID but not ITS. However, like Abadie et al. [2010, 2015], SHC restricts weights to the simplex. This guarantees interpretability but can be too restrictive when noise or complex dynamics make exact pre-treatment fit unattainable.

This paper introduces the Augmented Synthetic Historical Control method (ASHC), which extends SHC in two key ways. First, ASHC relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This permits negative weights, allowing controlled extrapolation beyond the convex hull of observed segments in exchange for better fit. Second, ASHC adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM [Ben-Michael et al., 2021]. This encourages the estimator to remain close to SHC unless extrapolation is needed. ASHC retains the stabilizing Gram penalty from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings where exact pre-treatment fit is otherwise impossible.

The paper makes three contributions. Methodologically, it introduces the ASHC estimator, which generalizes SHC by permitting limited, penalized extrapolation while maintaining the stabilizing Gram penalty. Theoretically, it derives finite-sample error bounds that decompose ASHC prediction error into four interpretable parts: the SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. It further establishes asymptotic consistency and a convergence rate of $O_p(m^{-H/(2H+3)})$, which approaches the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order H increases. Practically, this means that when the pre-treatment length m is large (e.g. $m = 120$) and H grows, the convergence rate becomes practically close to the parametric $O_p(m^{-1/2})$, so ASHC achieves efficient performance in realistic sample sizes.

While the finite-sample analysis provides guarantees on the accuracy of ASHC for realistic data sizes, asymptotic results are equally important. The finite-sample bounds show how ASHC predictions may differ from SHC depending on regularization strength, operator norm, smoothness bias, and noise. The asymptotic results complement this by demonstrating that, as the amount of pre-treatment data grows, ASHC's prediction error vanishes at a near-parametric rate under mild conditions. Together, these perspectives ensure that ASHC is both practically reliable in finite samples and theoretically sound in large-sample limits.

The discussion begins with a careful review of SHC, its assumptions, and its implementation, before introducing the ASHC framework. We then turn to the theoretical analysis, establishing finite-sample error bounds and asymptotic properties. The empirical performance of ASHC is explored through calibrated simulations, followed by an application that illustrates its usefulness in practice. Proofs are collected in the appendix, and all computations are carried out using Python's mlsynth library (https://mlsynth.readthedocs.io/en/latest/).

For policy analysts, the message is straightforward: ASHC provides a principled way to construct reliable counterfactuals even in environments where no external donor units are available. Furthermore, it allows policy analysts to do this even when methods such as interrupted time series method or the Chen et al. [2024] would not do well.

The remainder of the paper proceeds as follows. Section 1.1 situates our contribution within the existing literature on causal inference in policy evaluation. We then review the SHC method in Section 2.1 before introducing the augmented SHC (ASHC) in Section 2.2. Section 3 presents our theoretical results, including finite-sample error bounds and asymptotic properties. We then turn to an empirical application in Section 4, replicating the analysis of Barcelona's 2015 hotel moratorium from Bel et al. [2021], a setting that allows us to compare ITS, SCM, SHC, and ASHC on equal footing. Finally, Section 5 concludes with implications for both methodological research and policy analysis.

1.1 Literature Review

This paper is situated within the literature on causal time series methods. The most popular method is likely the Bayesian Structural Time Series method [Brodersen et al., 2015, **?**, **?**] due to its ease of use in Python and R as well as inferential methodology. Another similar estimator in the same vein is the causal ARIMA method [Menchetti et al., 2022].

2 Methodology

2.1 The Synthetic Historical Control Method

Let $\mathcal{T} = \{1, \dots, T\}$ index time, with pre-treatment period $\mathcal{T}_1 = \{t \in \mathcal{T} : t \leq T_0\}$ and post-treatment period $\mathcal{T}_2 = \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$, and an evaluation window of length m is defined as the final m pre-treatment periods $\mathcal{T}_{\text{eval}} = \{T_0 - m + 1, \dots, T_0\}$. Let $\mathbf{y} = (y_1, \dots, y_T)^\top$ denote the observed outcome series. The evaluation subvector is $\mathbf{y}_{\text{eval}} = \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector is $\mathbf{y}_{\text{post}} = \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

For each $t \in \mathcal{T}$, let y_t^1 and y_t^0 denote the potential outcomes under treatment and no treatment, respectively. The observed outcome is

$$y_t = d_t y_t^1 + (1 - d_t) y_t^0, \quad d_t = \mathbf{1}\{t > T_0\}.$$

For $t > T_0$, only y_t^1 is observed. SHC provides an estimate $\widehat{y}_t^0$ for the missing counterfactual, and the treatment effect is estimated as $\widehat{\alpha}_t = y_t - \widehat{y}_t^0, \quad t \in \mathcal{T}_2$, with the average treatment effect on the treated (ATT) defined as $\widehat{\text{ATT}} = n^{-1} \sum_{t \in \mathcal{T}_2} \widehat{\alpha}_t$. The SHC framework is grounded in the following assumptions.

Assumption 2.1 (Error Properties)**.** *The error term satisfies* $\mathbb{E}[\varepsilon_t] = 0$ *and* $\mathbb{E}[\varepsilon_t^2] < \infty$ *for all* $t \in \mathcal{T}$*.*

Assumption 2.2 (Latent Smoothness and Matching)**.** *The latent trend $\widetilde{y}(t)$ is $(H+1)$ times differentiable for some integer $H \geq 3$, with uniformly bounded derivative. Furthermore, there exists $\mathbf{w}^* \in \Delta^{N-1}$ such that $\mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*$, where Δ^{N-1} denotes the probability simplex.*

Assumption 2.3 (Feasibility and Stability)**.** *There exists a sparse weight vector $\mathbf{w} \in \Delta^{N-1}$ such that $\mathbf{y}_{eval} \approx \widetilde{\mathbf{Y}}_{pre}\mathbf{w}$ and the corresponding Gram matrix is positive definite.*

These assumptions ensure that noise is mean-zero and bounded, that the latent component is smooth enough to be locally approximated, and that the evaluation window can be matched stably from the donor pool of pre-treatment segments.

The implementation of SHC begins by smoothing the pre-treatment series $\mathbf{y}_{\text{pre}}$ using local linear regression with a Gaussian kernel, producing a smoothed trajectory $\widetilde{\mathbf{y}}$. From this smoothed series, N overlapping segments of length m are extracted to form the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$. The evaluation vector $\mathbf{y}_{\text{eval}}$ is reconstructed as a convex combination of the donor segments by solving the quadratic program

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}\mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2, \qquad \text{subject to } \mathbf{w} \geq \mathbf{0},\ \mathbf{1}^\top \mathbf{w} = 1,$$

where $\mathbf{C}_0$ contains the eigenvectors of the donor Gram matrix associated with eigenvalues below a threshold τ. The additional penalty shrinks weights along low-variance directions, mitigating instability from collinearity. Applying $\mathbf{w}^*$ to the aligned post-treatment donor matrix yields $\widehat{\mathbf{y}}^0_{\text{post}}$, which serves as the SHC estimate of the untreated counterfactual trajectory.

2.2 The Augmented Synthetic Historical Control Method

The augmented synthetic historical control (ASHC) method extends SHC by permitting limited extrapolation beyond the convex hull of historical segments, while stabilizing the solution through a proximity penalty. This modification is motivated by cases where the simplex constraint of SHC is too restrictive to achieve adequate pre-treatment fit. By allowing negative weights under an affine constraint and penalizing deviations from the SHC solution, ASHC balances improved fit with regularization.

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC weight vector. The ASHC estimator solves

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}\mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1.$$

As we can see, the first two terms are essentially unchanged. We have a fit term to learn the treated unit's outcomes in the evaluation period and the stability penalty term from Chen et al. [2024]. The third term in the ASHC objective,

$$\lambda \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|_2^2,$$

is a proximity penalty. This term discourages the ASHC solution from deviating too far from the original SHC weights. The penalty plays a dual role. Conceptually, it regularizes the affine extension of SHC by shrinking the solution back toward the purely convex combination used in SHC. This ensures that the additional flexibility provided by negative weights is only exercised when it leads to a substantial reduction in pre-treatment error. Mathematically, the penalty stabilizes the optimization problem by making the objective function strongly convex. The tuning parameter $\lambda \geq 0$ controls the degree of shrinkage: when λ is very large, the ASHC solution converges to $\mathbf{w}^{\text{SHC}}$, while when $\lambda = 0$ the penalty vanishes and the estimator reduces to an unpenalized affine extension of SHC. Thus, the proximity penalty creates a continuum between SHC and its affine relaxation, allowing researchers to balance interpretability and stability against the potential gains in fit from extrapolation.

The ASHC counterfactual is obtained as $\widehat{\mathbf{y}}^{0,\text{ASHC}}_{\text{post}} = \widetilde{\mathbf{Y}}_{\text{post}}\, \mathbf{w}^{\text{ASHC}}$, with treatment effects estimated analogously to SHC. In addition to the assumptions underlying SHC, ASHC requires bounds on the operator norms of the donor matrices.

2.3 Bandwidth and Regularization Parameter Selection

The performance of ASHC depends critically on two hyperparameters: the smoothing bandwidth h and the proximity penalty parameter λ. Both are chosen via cross-validation.

First I discuss the bandwidth for the kernel smoothing. Let $\mathbf{y}_{\text{pre}} = (y_1, \ldots, y_{T_0})^\top$ denote the pre-treatment series.
For a candidate bandwidth $h > 0$, define the local linear smoother estimate $\widehat{y}_i^{(-i)}(h)$ as the fitted intercept obtained by regressing $\{y_j : j \neq i\}$ on $\{j - i : j \neq i\}$ with kernel weights

$$K_h(j-i) \;=\; \exp\Bigl(-\tfrac{1}{2}\left(\tfrac{j-i}{h}\right)^2\Bigr),$$

normalized so that $\sum_{j \neq i} K_h(j-i) = 1$. Formally, for each $i \in \{1, \ldots, T_0\}$,

$$\widehat{\beta}(h)^{(-i)} = \arg\min_{\beta_0, \beta_1} \sum_{j \neq i} K_h(j-i) \Bigl(y_j - \beta_0 - \beta_1 (j-i)\Bigr)^2,$$

and the leave-one-out prediction is

$$\widehat{y}_i^{(-i)}(h) \;=\; \widehat{\beta}_0(h)^{(-i)}.$$

The LOOCV error is

$$\mathrm{CV}(h) \;=\; \frac{1}{T_0} \sum_{i=1}^{T_0} \Bigl(y_i - \widehat{y}_i^{(-i)}(h)\Bigr)^2.$$

The optimal bandwidth is then chosen as

$$h^* = \arg\min_{h \in \mathcal{H}} \mathrm{CV}(h),$$

where $\mathcal{H}$ is a prespecified candidate grid. This is part of the normal SHC method.

Next we discuss lambda, exclusive to the ASHC. Recall the ASHC objective

$$Q(\mathbf{w}; \lambda) = \Bigl\|\mathbf{y}_{\text{train}} - \widetilde{\mathbf{Y}}_{\text{train}} \mathbf{w}\Bigr\|_2^2 + \varsigma \left\|\mathbf{C}_0^\top \mathbf{w}\right\|_2^2 + \lambda \left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top \mathbf{w} = 1.$$

Let the evaluation window $\mathcal{T}_{\text{eval}}$ be split into a training set $\mathcal{T}_{\text{train}}$ and a validation set $\mathcal{T}_{\text{val}}$, with corresponding subvectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}$ and donor matrices $\widetilde{\mathbf{Y}}_{\text{train}}, \widetilde{\mathbf{Y}}_{\text{val}}$.

For each $\lambda > 0$,

$$\widehat{\mathbf{w}}(\lambda) = \arg\min_{\mathbf{w} \in \mathbb{R}^N} Q(\mathbf{w}; \lambda),$$

and the validation error is

$$\mathrm{Val}(\lambda) = \frac{1}{|\mathcal{T}_{\text{val}}|} \left\|\mathbf{y}_{\text{val}} - \widetilde{\mathbf{Y}}_{\text{val}}\, \widehat{\mathbf{w}}(\lambda)\right\|_2^2.$$

The optimal penalty parameter is selected as

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathrm{Val}(\lambda),$$

where Λ is a candidate grid of λ values (e.g. logarithmically spaced).

Together, h^* and λ^* balance the dual bias–variance tradeoffs of ASHC: smoothing bias versus noise, and extrapolation bias versus instability.

3 Theoretical Analysis of ASHC

One of the main concerns we had in mind while developing the ASHC estimator was the possibility of poor in-sample fit when using the standard SHC estimator. Thus, now that we have introduced the ASHC estimator, in this section we develop the finite-sample guarantees and asymptotic properties of the ASHC method. We work within the fixed N, large T asymptotic framework. After studying the deviation of the weights from ASHC relative to the SHC, we proceed to quantify the worst case in-sample error bound. Complete proofs are deferred to Appendix~**??**. We begin by stating two additional assumptions needed for the analysis.

Assumption 3.4 (Operator Norm Bound)**.** *The pre-treatment donor matrix satisfies* $\|\widetilde{\mathbf{Y}}\| \le \gamma$ *for some constant* $\gamma > 0$.

Assumption 3.5 (Post-Treatment Operator Norm)**.** *There exists* $\gamma_{post} > 0$ *such that* $\|\widetilde{\mathbf{Y}}_{post}\| \le \gamma_{post}$.

Assumption~3.4 ensures that the pre-treatment donor matrix is not "too large" in operator norm, preventing instability in the weight estimates under affine relaxation. Assumption~3.5 provides a parallel safeguard in the post-treatment period, ensuring that extrapolated counterfactuals do not blow up in variance.

3.1 Weight deviation from SHC

Let $\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}$ denote the deviation of ASHC weights from the SHC solution.

Lemma 3.1 (Deviation from SHC)**.** *Under Assumptions 2.1–3.4,*

$$\|\Delta\| \;\le\; \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\,\epsilon_{SHC},$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ *is the SHC in-sample fit error.*

Remark 3.1. *This lemma directly links the deviation in the weights,* $\|\Delta\|$*, to the change in the fitted donor outcomes,* $\|\widetilde{\mathbf{Y}}\Delta\|$*. It uses the operator norm of the donor matrix* $\widetilde{\mathbf{Y}}$*. The operator norm,* γ*, measures the maximum possible "stretching" effect that the donor matrix can have on any vector. It tells us that even if the ASHC weights differ from the SHC weights, the resulting change in the fitted donor outcomes is bounded. Specifically, it cannot exceed the magnitude of the weight deviation multiplied by* γ*.l.*

This is important because it ensures that the model is well-behaved. If the donor matrix had a very large operator norm, a small change in the weights could lead to a huge, uncontrolled change in the fitted outcomes. By bounding this norm with γ, the lemma guarantees that the impact of weight deviations on the fitted outcomes is predictable and proportional.

3.2 Finite-sample error bounds

Using Lemma~3.1, we can bound ASHC's in-sample error relative to SHC.

Proposition 3.1 (Pre-treatment Error Bound)**.** *Under Assumptions 2.1–2.3,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \;\le\; \epsilon_{SHC}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

Remark 3.2. *The proposition explicitly shows how each component contributes to the final bound: (1) the SHC baseline error, (2) the ASHC deviation penalty controlled by* λ *and* ς*, (3) the smoothness bias from Taylor approximation, and (4) the bounded noise. Factoring* ϵ_{SHC} *at the end highlights how ASHC scales the SHC error by at most a small multiplicative factor depending on the regularization strength. It is worth emphasizing that the ASHC estimator is guaranteed to perform at least as well as SHC within the evaluation window, since its objective explicitly penalizes deviations from the SHC solution. The bound above quantifies how much its predictions may differ, depending on the donor matrix and the penalty strength. However, this guarantee applies in-sample. Out-of-sample performance in the post-treatment period depends on smoothness of the latent trend and the boundedness of noise, as reflected in the* $b_{\epsilon(H,k)}$ *and* δ *terms.*

Proposition 3.2 (Out-of-sample Error Bound)**.** *Under Assumptions 2.1–3.5,*

$$\left\|\mathbf{y}_{post} - \mathbf{Y}_{post}\mathbf{w}^{ASHC}\right\| \le b_{\epsilon,post}(H,k) + \delta_{post} + \gamma_{post}\left(\eta + \frac{\epsilon_{SHC}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right),$$

where η bounds the difference $\|\mathbf{w}^* - \mathbf{w}^{SHC}\|$.

Remark 3.3. *Proposition 3.2 underscores that the accuracy of ASHC's post-treatment predictions, and therefore the ATT estimates, depends on both model fit and regularization. The smoothness bias $b_{\epsilon,post}(H,k)$ and noise δ_{post} are unavoidable statistical components. The term involving ϵ_{SHC} shows that if SHC fits poorly, ASHC can reduce residual error, but the regularization parameter λ ensures this improvement is controlled and stable. The constant η reflects the gap between SHC and the infeasible oracle weights, capturing how much better one could hope to do with perfect knowledge. For applied researchers, this result highlights a key trade-off: ASHC will not dramatically diverge from SHC unless SHC itself leaves a poor fit, and even then, its adjustments are bounded by both regularization and operator norm constraints. In practice, this means that adopting ASHC provides a safeguard: one gains the possibility of better fit without opening the door to highly unstable or implausible counterfactual estimates.*

This inequality highlights what governs the reliability of ASHC's counterfactuals: smoothness bias, outcome noise, the quality of SHC's fit, and the penalty parameter λ.

3.3 Asymptotic properties

We next examine large-sample behavior.

Theorem 3.1 (Consistency of ASHC)**.** *Under Assumptions 2.1–3.5, as $m \to \infty$ and $H \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \longrightarrow \delta.$$

If $\boldsymbol{\eta}$ is sub-Gaussian, then

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \xrightarrow{p} 0.$$

This result shows that with enough pre-treatment data and smooth underlying trends, ASHC's prediction error shrinks to the noise level, and under standard probabilistic assumptions it converges in probability to zero. In practice, this ensures that with long enough evaluation windows, ASHC delivers consistent counterfactuals.

Corollary 3.1 (Convergence rate)**.** *The ASHC estimator is asymptotically unbiased up to δ. If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| = O_p\Big(m^{-H/(2H+3)}\Big).$$

For large H, this approaches the parametric rate $O_p(m^{-1/2})$.

Remark 3.4. *This convergence rate shows how quickly ASHC's error vanishes as the evaluation window length m increases. The rate depends on the smoothness parameter H: with more smoothness, the error decreases faster. When H is large, the rate approaches the familiar $O_p(m^{-1/2})$ seen in parametric settings, meaning ASHC becomes nearly as efficient as standard regression-type estimators.*

ASHC combines flexibility in finite samples with strong long-run efficiency guarantees, similar to the bias bound developed by Abadie et al. [2010] for the SC estimator. This result underscores the value of longer pre-treatment windows: more data allow ASHC to exploit smoothness and approach parametric efficiency, while shorter pre-treatment windows will yield slower convergence typical of nonparametric methods. In practice, this means that if sufficient pre-treatment data are available, ASHC provides consistent and statistically unbiased estimates.

These results carry three main lessons. First, ASHC remains close to SHC when SHC already fits well, so adopting ASHC does not risk worse in-sample fit. Second, when SHC leaves residual error, ASHC can reduce that error, but regularization ensures the counterfactual does not become unstable. Third, with sufficient data, ASHC is consistent and approaches parametric efficiency, making it a reliable tool for long evaluation windows. For the applied microeconomist or policy analyst, the method offers a practical compromise: better pre-treatment balance where needed, without losing the interpretability and stability of the SHC framework, thereby allowing researchers to estimate counterfactuals even without donor units or perfect fit from convex regression estimators.

4 Empirical Application

We replicate the setting of Bel et al. [2021], who study the effects of Barcelona's 2015 hotel moratorium using data on hotel prices from 2010 to 2017. This application provides a compelling case to benchmark ITS, SCM, SHC, and ASHC against one another.

5 Conclusion

Appendix

Technical Proofs

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A Useful Lemmas

Proof of Lemma 3.1. To analyze the ASHC optimization problem, recall the objective function:

$$J(\mathbf{w}) = \frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2, \tag{1.1}$$

subject to the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$.

To enforce this constraint, we form th Lagrangian:

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1), \tag{1.2}$$

where μ is a Lagrange multiplier ensuring feasibility of $\mathbf{w}$. We compute the gradient of $\mathcal{L}$ with respect to $\mathbf{w}$ by differentiating each term separately.

For the first term, define the residual vector:

$$r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}. \tag{1.3}$$

The fit term is then:

$$\frac{1}{2\lambda}\|r(\mathbf{w})\|^2 = \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w}). \tag{1.4}$$

We can see that this is a standard quadratic form. We begin by applying the Chain Rule for vector calculus. The chain rule for a composite function $f(g(\mathbf{w}))$ is given by:

$$\nabla_{\mathbf{w}} f(g(\mathbf{w})) = \left(\nabla_{g(\mathbf{w})} f\right)^\top \nabla_{\mathbf{w}} g(\mathbf{w})$$

In our case, the inner function is the residual vector $g(\mathbf{w}) = r(\mathbf{w})$, and the outer function is the squared norm $f(r) = \frac{1}{2\lambda} r^\top r$. First, let's find the gradient of the outer function with respect to the residual vector r.

$$\nabla_r \left(\frac{1}{2\lambda} r^\top r\right) = \frac{1}{2\lambda}\nabla_r \left(\sum_i r_i^2\right) = \frac{1}{2\lambda}\begin{pmatrix} 2r_1 \\ 2r_2 \\ \vdots \end{pmatrix} = \frac{1}{\lambda} r$$

This is a well-known result: the gradient of $\|r\|^2$ is $2r$. The $\frac{1}{2\lambda}$ is a constant scaling factor.

Next, we need to find the gradient of the inner function, $r(\mathbf{w})$, with respect to the weight vector $\mathbf{w}$. This is the Jacobian matrix of the transformation.

$$\nabla_{\mathbf{w}} r(\mathbf{w}) = \nabla_{\mathbf{w}}(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Since $\widetilde{\mathbf{y}}$ is a constant vector with respect to $\mathbf{w}$, its gradient is zero. We only need to compute the gradient of $-\widetilde{\mathbf{Y}}\mathbf{w}$.

$$\nabla_{\mathbf{w}}(-\widetilde{\mathbf{Y}}\mathbf{w}) = -\widetilde{\mathbf{Y}}$$

This result comes from the general rule that for a linear function $A\mathbf{x}$, the gradient with respect to $\mathbf{x}$ is just the matrix A. Here, our linear function is a negative, so the gradient is $-\widetilde{\mathbf{Y}}$.

Now we plug the results from the above back into the chain rule formula.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\nabla_{r(\mathbf{w})} \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w})\right)^\top \nabla_{\mathbf{w}} r(\mathbf{w})$$

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\frac{1}{\lambda} r(\mathbf{w})\right)^\top (-\widetilde{\mathbf{Y}})$$

Using the property that $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$, we get:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda} r(\mathbf{w})^\top (-\widetilde{\mathbf{Y}}) = -\frac{1}{\lambda} r(\mathbf{w})^\top \widetilde{\mathbf{Y}}$$

This is equivalent to:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top r(\mathbf{w})$$

Finally, we substitute the definition of the residual vector, $r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}$, back into the gradient expression.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Distributing the negative sign gives us the final form:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}})$$

Next, consider the deviation penalty term:

$$\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = (\mathbf{w} - \mathbf{w}^{\mathrm{SHC}})^\top(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.5}$$

Its gradient with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}}\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.6}$$

Notice that the Hessian of this term is $2\mathbf{I}$, because differentiating again with respect to $\mathbf{w}$ gives:

$$\nabla^2_{\mathbf{w}}\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2\mathbf{I}. \tag{1.7}$$

Here $\mathbf{I}$ is the identity matrix, representing curvature equally in all coordinate directions. Thus the $2\mathbf{I}$ in M directly reflects the Hessian of this quadratic deviation penalty.

For the Gram penalty term, rewrite it as:

$$\varsigma\|\mathbf{C}_0^\top\mathbf{w}\|^2 = \varsigma\mathbf{w}^\top\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}. \tag{1.8}$$

Since $\mathbf{C}_0\mathbf{C}_0^\top$ is symmetric, its gradient is:

$$\nabla_{\mathbf{w}}\varsigma\mathbf{w}^\top\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} = 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}. \tag{1.9}$$

Finally, the affine constraint term differentiates as:

$$\nabla_{\mathbf{w}}\mu(\mathbf{1}^\top\mathbf{w} - 1) = \mu\mathbf{1}. \tag{1.10}$$

We now have the first-order condition from the gradient of the Lagrangian:

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}}) + 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} + \mu\mathbf{1} = \mathbf{0}. \tag{1.11}$$

Expanding the deviation penalty term gives

$$2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) = 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}}. \tag{1.12}$$

Substituting back, the condition becomes

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w} - \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} + \mu\mathbf{1} = \mathbf{0}. \tag{1.13}$$

Now collect all the terms involving $\mathbf{w}$ on the left-hand side:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} \tag{1.14}$$

and move all the other terms to the right-hand side:

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.15}$$

Thus we obtain the compact matrix form:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.16}$$

Here the $2\mathbf{I}$ arises directly from the quadratic penalty $\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2$, whose gradient contributes $2\mathbf{w}$, corresponding in matrix form to $2\mathbf{I}\cdot\mathbf{w}$.

Next, define the matrix

$$M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top. \tag{1.17}$$

M corresponds to the Hessian of the objective function $J(\mathbf{w})$ plus the identity-weighted contribution of the deviation penalty.

The solution $\mathbf{w}^{\mathrm{ASHC}}$ satisfies the first-order condition:

$$M\mathbf{w}^{\mathrm{ASHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.18}$$

Evaluating the same matrix M at $\mathbf{w}^{\mathrm{SHC}}$, which is not necessarily the ASHC solution, gives:

$$M\mathbf{w}^{\mathrm{SHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}. \tag{1.19}$$

Subtracting this equation from the previous one, we get

$$M\mathbf{w}^{\mathrm{ASHC}} - M\mathbf{w}^{\mathrm{SHC}} = \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}\right) - \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}\right). \tag{1.20}$$

Simplifying the right-hand side by canceling the $2\mathbf{w}^{\mathrm{SHC}}$ terms, we obtain

$$M(\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}) = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\left(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\right) - \mu\mathbf{1}. \tag{1.21}$$

Define the deviation

$$\Delta = \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}, \tag{1.22}$$

and the residual vector

$$r = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}. \tag{1.23}$$

Since both $\mathbf{w}^{\mathrm{ASHC}}$ and $\mathbf{w}^{\mathrm{SHC}}$ satisfy the affine constraint $\mathbf{1}^\top\mathbf{w} = 1$, their difference Δ lies in the hyperplane orthogonal to $\mathbf{1}$:

$$\mathbf{1}^\top\Delta = 0. \tag{1.24}$$

Taking the inner product of both sides of the previous equation with Δ gives

$$\Delta^\top M\Delta = \frac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r - \mu\underbrace{\Delta^\top\mathbf{1}}_{=0} = \frac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r. \tag{1.25}$$

Here, the right-hand side is an inner product between the deviation vector Δ and the vector $\widetilde{\mathbf{Y}}^\top r$. Because inner products can be difficult to control directly, we apply the Cauchy–Schwarz inequality to bound it:

$$|\Delta^\top\widetilde{\mathbf{Y}}^\top r| \le \|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|. \tag{1.26}$$

Thus, $\Delta^\top M\Delta \leq \frac{1}{\lambda}\|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|$. By assumption, the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, and the residual norm is $\epsilon_{\text{SHC}} = \|r\|$, so

$$\|\widetilde{\mathbf{Y}}^\top r\| \leq \|\widetilde{\mathbf{Y}}\|\|r\| \leq \gamma\epsilon_{\text{SHC}}. \tag{1.27}$$

Thus, $\Delta^\top M\Delta \leq \frac{\gamma}{\lambda}\|\Delta\|\epsilon_{\text{SHC}}$. On the other hand, since M is positive definite, its smallest eigenvalue $\lambda_{\min}(M)$ guarantees $\Delta^\top M\Delta \geq \lambda_{\min}(M)\|\Delta\|^2$. Putting these two inequalities together, we sandwich the quadratic form: $\lambda_{\min}(M)\|\Delta\|^2 \leq \Delta^\top M\Delta \leq \frac{\gamma}{\lambda}\|\Delta\|\,\epsilon_{\text{SHC}}$.

The left-hand side is the lower bound derived from the smallest eigenvalue of the positive definite matrix M. The right-hand side is the upper bound derived from the Cauchy-Schwarz and operator norm inequalities. We can now discard the middle term to focus on the key inequality between the two bounds: $\lambda_{\min}(M)\|\Delta\|^2 \leq \frac{\gamma}{\lambda}\|\Delta\|\,\epsilon_{\text{SHC}}$. Assuming $\|\Delta\| > 0$ (the trivial case where the weights are identical is already clear), we can divide both sides of the inequality by $\|\Delta\|$ to isolate the term of interest. $\lambda_{\min}(M)\|\Delta\| \leq \frac{\gamma}{\lambda}\epsilon_{\text{SHC}}$. Rearranging this inequality to solve for $\|\Delta\|$ gives us an intermediate bound: $\|\Delta\| \leq \frac{\gamma}{\lambda\lambda_{\min}(M)}\epsilon_{\text{SHC}}$. Next, we establish a lower bound for the smallest eigenvalue of the matrix M. Recall that the matrix M is defined as: $M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top$. The smallest eigenvalue of a sum of positive semi-definite matrices is at least the sum of their smallest eigenvalues. Since the first term's smallest eigenvalue is non-negative, and the second term's is 2, we have:

$$\lambda_{\min}(M) \geq \lambda_{\min}\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\right) + \lambda_{\min}(2\mathbf{I}) + \lambda_{\min}(2\varsigma\mathbf{C}_0\mathbf{C}_0^\top)$$

$$\lambda_{\min}(M) \geq 0 + 2 + 2\varsigma\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top)$$

By the assumption that $\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top) \geq \|\mathbf{C}_0\|^2$, we get the lower bound for the smallest eigenvalue of M:

$$\lambda_{\min}(M) \geq 2 + 2\varsigma\|\mathbf{C}_0\|^2$$

Finally, we substitute this lower bound for $\lambda_{\min}(M)$ back into the intermediate bound for $\|\Delta\|$. A smaller value in the denominator leads to a larger (and therefore valid) upper bound on the expression. This gives us the final bound on the deviation:

$$\|\Delta\| \leq \frac{\gamma}{\lambda(2 + 2\varsigma\|\mathbf{C}_0\|^2)}\epsilon_{\text{SHC}} = \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\epsilon_{\text{SHC}}$$

This final expression shows that the deviation between the ASHC and SHC weights is controlled by the SHC residual error, scaled by a factor that depends on the regularization parameters λ and ς. ☐

Lemma 1.2. *Under Assumption (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|,$$

where $\Delta = \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$.

Proof of Lemma 1.2. We begin by recalling the definition of the operator (spectral) norm of a matrix. For a real matrix $A \in \mathbb{R}^{m\times N}$,

$$\|A\| := \sup_{\mathbf{v}\in\mathbb{R}^N\setminus\{0\}} \frac{\|A\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This means $\|A\|$ measures the largest possible multiplicative factor by which A can increase the Euclidean norm of a vector.

Now let $A = \widetilde{\mathbf{Y}}$ and $\mathbf{v} = \Delta$, where $\Delta = \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}} \in \mathbb{R}^N$ is the deviation of the ASHC weight vector from the SHC weight vector.

By the definition of the operator norm, we have

$$\frac{\|\widetilde{\mathbf{Y}}\Delta\|}{\|\Delta\|} \leq \|\widetilde{\mathbf{Y}}\|, \quad \text{for all } \Delta \neq \mathbf{0}.$$

Multiplying through by $\|\Delta\|$ yields

$$\|\widetilde{\mathbf{Y}}\Delta\| \ \leq \ \|\widetilde{\mathbf{Y}}\| \, \|\Delta\|.$$

Finally, Assumption (A4) gives us the explicit bound

$$\|\widetilde{\mathbf{Y}}\| \ \leq \ \gamma,$$

so that

$$\|\widetilde{\mathbf{Y}}\Delta\| \ \leq \ \gamma\|\Delta\|.$$

In words: although ASHC may choose weights that differ from SHC, the resulting perturbation in the reconstructed donor outcomes is controlled in magnitude, with the constant of proportionality γ coming directly from the operator norm bound in Assumption (A4). □

Lemma 1.3. *Under (A3), for horizon $k > 0$,*

$$|\ell(T_0 + k) - \ell^{SHC}(T_0 + k)| \ \leq \ b_\epsilon(H, k) \ := \ \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof of Lemma 1.3. See Proposition 1 of Chen et al. [2024]. □

Remark 1.5. *Lemma 1.3 formalizes the idea that the SHC predictor inherits the smoothness of the latent trend $\ell(\cdot)$. Because SHC is built from overlapping donor segments that reflect the same underlying smooth dynamics, the approximation reproduces the first H terms of the Taylor expansion around T_0. The only discrepancy comes from the $(H+1)$-st order remainder, which is small when the latent trend is sufficiently smooth. Thus, the difference between ℓ and ℓ^{SHC} grows only polynomially in the forecast horizon k, and the error is of order $|k|^{H+1}$ rather than linear or quadratic in k. This ensures that SHC extrapolates the smooth component accurately over short to moderate horizons.*

Lemma 1.4 (Post-treatment smoothness bias and noise)**.** *Under Assumptions (A2) and (A3), the approximation error in the post-treatment period satisfies*

$$\|\mathbf{y}_{post} - \mathbf{Y}_{post}\mathbf{w}^*\| \leq b_{\epsilon,post}(H, k) + \delta_{post},$$

where $b_{\epsilon,post}(H, k)$ is a smoothness bias term similar to the pre-treatment case, and δ_{post} bounds the noise in the post-treatment outcomes.

Proof of Lemma 1.4. Recall from Assumption (A1) that the treated unit's latent potential outcome $\ell(t)$ is $H + 1$ times differentiable, and its $(H + 1)$-th derivative is bounded by ϵ. By a Taylor expansion argument, the latent outcome over the post-treatment period can be approximated by the linear combination of donor segments with a bias bounded by the smoothness term $b_{\epsilon,\text{post}}(H, k)$.

Additionally, from Assumption (A2), the noise vector $\boldsymbol{\eta}_{\text{post}}$ satisfies $\|\boldsymbol{\eta}_{\text{post}}\| \leq \delta_{\text{post}}$.

Thus,

$$\mathbf{y}_{\text{post}} = \mathbf{Y}_{\text{post}}\mathbf{w}^* + \boldsymbol{\eta}_{\text{post}} + \text{smoothness bias},$$

where the smoothness bias norm is at most $b_{\epsilon,\text{post}}(H, k)$.

Applying the triangle inequality,

$$\|\mathbf{y}_{\text{post}} - \mathbf{Y}_{\text{post}}\mathbf{w}^*\| \leq \|\boldsymbol{\eta}_{\text{post}}\| + b_{\epsilon,\text{post}}(H, k) \leq \delta_{\text{post}} + b_{\epsilon,\text{post}}(H, k).$$

This completes the proof. □

Lemma 1.5 (Weight deviation between oracle and SHC)**.** *Under Assumptions (A1)–(A5), the difference between the oracle weights* $\mathbf{w}^*$ *and the SHC weights* $\mathbf{w}^{SHC}$ *is bounded by*

$$\|\mathbf{w}^* - \mathbf{w}^{SHC}\| \leq \eta,$$

for some $\eta > 0$ *depending on the data and model.*

Proof of Lemma 1.5. This follows from the stability and consistency properties of the SHC estimator under Assumptions (A1)–(A5). Since $\mathbf{w}^{\mathrm{SHC}}$ solves a constrained least squares problem minimizing

$$\left\|\widetilde{\mathbf{y}}_{\mathrm{eval}} - \widetilde{\mathbf{Y}}_{\mathrm{eval}}\mathbf{w}\right\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2,$$

and $\mathbf{w}^*$ satisfies the noiseless model $\widetilde{\mathbf{y}}_{\mathrm{eval}} = \widetilde{\mathbf{Y}}_{\mathrm{eval}}\mathbf{w}^* + \boldsymbol{\eta}$ with $\|\boldsymbol{\eta}\| \leq \delta$, standard results on constrained least squares imply $\mathbf{w}^{\mathrm{SHC}}$ is close to $\mathbf{w}^*$ within a ball whose radius depends on δ and model parameters.

More explicitly, since the problem is strictly convex and well-conditioned by assumption (due to the regularization and eigenvalue structure),

$$\|\mathbf{w}^* - \mathbf{w}^{\mathrm{SHC}}\| \leq \eta,$$

where η depends on the noise bound δ, the regularization parameter ς, and the spectral properties of $\widetilde{\mathbf{Y}}_{\mathrm{eval}}$.

A detailed exact expression can be derived by comparing the KKT conditions for $\mathbf{w}^*$ and $\mathbf{w}^{\mathrm{SHC}}$, but for the purposes here, it suffices to assert the existence of such η. □

Lemma 1.6 (Total weight deviation bound)**.** *We have*

$$\|\mathbf{w}^* - \mathbf{w}^{ASHC}\| \leq \|\mathbf{w}^* - \mathbf{w}^{SHC}\| + \|\mathbf{w}^{SHC} - \mathbf{w}^{ASHC}\|.$$

Proof of Lemma 1.6. This is a direct application of the triangle inequality for vector norms. For any vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in a normed vector space,

$$\|\mathbf{a} - \mathbf{c}\| \leq \|\mathbf{a} - \mathbf{b}\| + \|\mathbf{b} - \mathbf{c}\|.$$

Setting $\mathbf{a} = \mathbf{w}^*$, $\mathbf{b} = \mathbf{w}^{\mathrm{SHC}}$, and $\mathbf{c} = \mathbf{w}^{\mathrm{ASHC}}$ yields the stated result. □

B Propositions

Proof of Proposition 3.1. We want to control the error of the ASHC estimator, given by

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

Directly analyzing this is difficult because the weights $\mathbf{w}^{\mathrm{ASHC}}$ are defined through a penalized optimization problem. To make progress, we first recall the triangle inequality, which states that for any vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|.$$

This allows us to introduce the SHC weights into the expression and split the ASHC error into two pieces: the SHC prediction error, which we already know how to bound, and the extra error due to the deviation of ASHC from SHC.

Let $\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}$ denote the difference between the two weight vectors. Then we can write

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| = \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + \widetilde{\mathbf{Y}}(\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}})\|.$$

Applying the triangle inequality gives

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \ \leq \ \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\| + \|\widetilde{\mathbf{Y}}(\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}})\|.$$

Since $\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}} = -\Delta$, the second term becomes $\|\widetilde{\mathbf{Y}}\Delta\|$. Denoting the SHC error by

$$\epsilon_{\mathrm{SHC}} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\|,$$

we obtain

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \ \leq \ \epsilon_{\mathrm{SHC}} + \|\widetilde{\mathbf{Y}}\Delta\|.$$

The challenge now reduces to bounding $\|\widetilde{\mathbf{Y}}\Delta\|$. For this, we use the operator norm. Recall that for a matrix A,

$$\|A\| := \sup_{\mathbf{v}\neq 0} \frac{\|A\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This definition guarantees that for any vector $\mathbf{v}$,

$$\|A\mathbf{v}\| \leq \|A\| \cdot \|\mathbf{v}\|.$$

Taking $A = \widetilde{\mathbf{Y}}$ and $\mathbf{v} = \Delta$, we have

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \|\widetilde{\mathbf{Y}}\| \cdot \|\Delta\|.$$

By Assumption (A4), the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, so

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|.$$

To proceed, we need a bound on the size of the deviation Δ. Lemma 3.1 provides exactly this, showing that

$$\|\Delta\| \leq \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Substituting this into the previous inequality, we obtain

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Finally, we incorporate the other two sources of error. Lemma 1.3 bounds the bias due to imperfect smoothness of the latent trend by

$$b_\epsilon(H, k) = \frac{2\epsilon|k|^{H+1}}{(H+1)!},$$

and Assumption (A2) ensures that the noise term contributes at most δ. Adding these yields

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} + b_\epsilon(H,k) + \delta.$$

It remains only to factor out ϵ_{SHC} from the first two terms:

$$\epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} = \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right).$$

Thus we conclude that

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta,$$

which completes the proof. □

Proof of Proposition 3.2. We want to bound the out-of-sample prediction error

$$\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\|.$$

Add and subtract the oracle weights prediction $\mathbf{Y}_{\mathrm{post}}\mathbf{w}^*$ inside the norm:

$$\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\| = \left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^* + \mathbf{Y}_{\mathrm{post}}\mathbf{w}^* - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\|.$$

By the triangle inequality,

$$\leq \underbrace{\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^*\right\|}_{\text{smoothness bias + noise}} + \underbrace{\left\|\mathbf{Y}_{\mathrm{post}}\left(\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}}\right)\right\|}_{\text{weight deviation effect}}.$$

Using Lemma 1.4, the first term is bounded by

$$\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^*\| \leq b_{\epsilon,\mathrm{post}}(H,k) + \delta_{\mathrm{post}}.$$

For the second term, apply the operator norm bound from Assumption 3.5:

$$\left\|\mathbf{Y}_{\mathrm{post}}(\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}})\right\| \leq \gamma_{\mathrm{post}}\left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}}\right\|.$$

By Lemma 1.6, decompose the weight difference:

$$\left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}}\right\| \leq \left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{SHC}}\right\| + \left\|\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}}\right\|.$$

Use Lemma 1.5 and your existing bound on $\|\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}\|$ (from Lemma 3.1 or your prior proof):

$$\leq \eta + \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Combining all parts,

$$\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\| \leq b_{\epsilon,\mathrm{post}}(H,k) + \delta_{\mathrm{post}} + \gamma_{\mathrm{post}}\left(\eta + \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right).$$

To bound the ATT estimation error, recall

$$\widehat{\mathrm{ATT}} - \mathrm{ATT} = \frac{1}{n}\sum_{t\in\mathcal{T}_2}\left(y_t(0) - \hat{y}_t(0)\right).$$

Applying Jensen's inequality and Cauchy–Schwarz,

$$\left|\widehat{\mathrm{ATT}} - \mathrm{ATT}\right| \leq \frac{1}{n} \sum_{t \in \mathcal{T}_2} |y_t(0) - \hat{y}_t(0)| \leq \frac{1}{\sqrt{n}} \left\| \mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}} \mathbf{w}^{\mathrm{ASHC}} \right\|.$$

Substitute the bound for the prediction error to complete the ATT bound. □

Proof of Corollary ??. Starting from Proposition 3.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Behavior of ϵ_{SHC}. By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\mathrm{SHC}} \to 0 \quad \text{as } m \to \infty,\ H \to \infty.$$

Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\mathrm{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}} \epsilon_{\mathrm{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Thus

$$b_\epsilon(H,k) \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \leq \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2, then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m \to \infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \leq \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. □

Proof of Corollary 3.1. Part 1: Consistency. From Theorem 3.1, the error converges to δ, or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 3.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Bias term. From Lemma 1.3,

$$b_\epsilon(H,k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p\big((mh_m)^{-1/2}\big).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\mathrm{SHC}} = O_p\big(h_m^H + (mh_m)^{-1/2}\big).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\mathrm{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$

Thus the total error is

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

Step 6. Near-parametric rate. As $H \to \infty$,

$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. $\square$

Remark 2.6. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \le \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

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